COURSE 2A

C2-09 CAPSTONE, PROMOTION & READINESS

CORE CAP1-CAP14 TEACHING BOOK - RC1

Meridian decision contract -> trusted analysis -> executive defense -> truthful career evidence

THIS BOOK IS YOUR TEACHER

PL-300 is a separate supplementary book and never blocks Course 2A completion or the direct C2B route.

This book is your teacher

Use one lesson at a time. New advanced concepts are still taught from first principles. Save a genuine attempt before opening protected review; do not skim ahead just because a topic name looks familiar.

Daily lesson loop

1) Why it matters -> 2) what you learn -> 3) prerequisite -> 4) picture it in your head -> 5) new words -> 6) watch / visual source when useful -> 7) follow with me once -> 8) expected result -> 9) why it worked -> 10) common beginner mistakes -> 11) recovery ladder -> 12) small fresh task -> 13) validate -> 14) teach it back -> 15) protected review + changed retry -> 16) evidence / mastery + exact next route.

Source rule

External videos are companions, not the course authority. If a useful source exists, open the exact link and return to the course task. If an external source would leak a solved capstone or add little value, the lesson explicitly uses an INTERNAL TEACHING ROUTE.

Attempt protection

Do the teacher example once. Then close it and complete the changed task from your own blank workbook/query/report/notebook. Protected review opens only after a saved attempt and is followed by a changed retry.

Mastery standard

You are finished only when you can reproduce the result, validate it, explain why it works, recover from a mistake and save the required evidence.

Contents / Index

Every canonical CAP lesson and actual page number

Prerequisite, tool and outcome map

Use the tools you already learned; do not add technology for prestige

ASSUMED

C2-01 Power Query; C2-02 SQL; C2-03 modeling; C2-04 DAX/reporting; C2-05 Python/reproducibility; C2-06 warehousing/quality; C2-07 enterprise operations; C2-08 stakeholder/communication. Any weak prerequisite is repaired locally.

REQUIRED ARTIFACTS

Meridian raw ZIP, Capstone Evidence Workbook, your SQL/Power Query/Python/PBIX or explicit design equivalents, validation log, executive brief, handover package, interview/promotional evidence, Final Gate A.

ENVIRONMENT BOUNDARIES

PostgreSQL and Power BI are strongly recommended for hands-on portfolio quality. If a specific live Service/Fabric feature is unavailable, record ENVIRONMENT BLOCKED; do not fake tenant evidence. Local project proof is not production proof.

OUTCOME BOUNDARY

C2-09 core completion proves this optional C2A route was completed under evidence rules. It does not guarantee employment, title, salary, Microsoft certification, or Data Engineer job readiness.

Capstone / review / Final Gate route

Attempt-before-review and fresh assessment remain non-negotiable

CAP1 contract -> CAP2-5 build -> CAP6-8 trust -> CAP9-10 handover -> CAP11-14 career -> Final Gate A

CAP ROUTE

Complete CAP1 through CAP14 in order because later evidence depends on earlier trusted artifacts. The protected capstone review is criterion-based and opens only after the matching CAP deliverable.

FINAL GATE A

Fresh Subscription Services case. It is not Meridian repeated. Closed-book/independent first attempt integrates requirements, grains, SQL/model/DAX, automation, quality, enterprise operations and executive defense.

RETEST B/C

Healthcare Network Operations and Logistics Marketplace remain hidden. Use only after targeted remediation. Changed data/process/incident conditions prevent answer memorization.

PL-300 SEPARATION

Optional certification route lives in C2_09_PL300_SUPPLEMENT_TEACHING_BOOK_RC1.pdf and has its own assessment/review/readiness gate. It never substitutes for CAP/Final Gate.

CAP1 - Requirements: receive and clarify an ambiguous business problem

1. Why it matters

The Meridian executive request is deliberately broad: 'Tell us where growth is healthy, where it is fragile, and what we should do next quarter.' Starting SQL immediately would force you to invent what healthy, fragile and next-quarter action mean.

2. What you are learning

Convert an ambiguous executive request into a one-page decision contract covering decision owner, audience, comparison period, scope, KPI definitions, required grains, data/freshness assumptions, deadline, risks, open questions and acceptance criteria.

3. Prerequisite / what you already know

SBA1 discovery, SBA4 evidence language and C2-00 grain/definition habits.

4. Picture it in your head

Mental model

Imagine this as a visible path: Executive request -> Decision -> Definitions -> Scope -> Questions -> Acceptance. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Decision contract: Written agreement on decision, definitions, scope, evidence and acceptance before analysis. Metric contract: Definition, formula/grain/time/filter rules and owner for a KPI. Acceptance criterion: Observable test showing the analytical deliverable is fit for the agreed decision.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Write the decision: leadership must choose next-quarter actions to protect sustainable growth, not simply rank regions.
2. Define candidate meanings for healthy growth: revenue trend, gross margin, target attainment, acquisition efficiency, customer health and service pressure. Mark which definitions require owner confirmation.
3. Set comparison period and grains. Sales is invoice-line, marketing Month-Region-Channel, targets Month-Region, health Customer-Month, support Ticket.
4. List open questions: Is revenue booked or recognized? What threshold makes growth fragile? Which actions are actually available next quarter? Who approves a metric conflict?
5. Write acceptance: key figures reconcile to raw controls; facts are not multiplied; observations are separated from causal claims; recommendation names owner/timing/limitation.

8. What you should see / be able to say

Expected result

A one-page brief that constrains later SQL/model/report decisions and exposes unresolved definitions instead of hiding them.

9. Why it worked

Reasoning

You created an analytical contract before calculations, reducing silent assumptions and rework. Professional upgrade: In real work, requirements remain living contracts: material scope/definition changes become decision-log entries and may require revalidation. Do not freeze ambiguity into code.

10. Common beginner mistakes

1. Treat executive wording as a complete requirement. | 2. Define fragile only after seeing which region looks bad. | 3. Choose chart/KPI set before decision owner and scope. | 4. Use one grain statement for all source files.

11. If you get stuck - recovery ladder

Recovery

Use six blanks: decision, owner/audience, KPI definitions, grains/period, unknowns/risks, acceptance. Do not calculate until all six are visible. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Write the full Meridian decision brief and identify at least five questions that require stakeholder confirmation before a production decision.

13. Validate your result

Validation

A reviewer can say what decision the analysis supports, what one row means in each source, which KPI definitions are provisional and what would make the work unacceptable.

14. Teach it back in your own words

Explain without reading

Explain why a precise SQL query can still answer the wrong business question.

15. Protected review + changed fresh retry

Attempt protection

Save the first decision brief. Protected review checks completeness only. Changed retry: a CFO reframes the request to 'protect margin even if revenue growth slows'. Update only the contract sections affected.

16. Evidence / mastery + exact next route

Mastery evidence

Decision brief + metric-definition table + open-question/owner log + acceptance checklist + explain-back. Use C2_09_MERIDIAN_CAPSTONE_DATA_RC1.zip README/source_contracts only; do not profile business values yet. Then continue to CAP2.

CAP2 - Data acquisition: determine required sources

1. Why it matters

The capstone contains several facts at different grains. The biggest trap is to join everything because the keys appear compatible. Source selection is a reasoning task before it is a loading task.

2. What you are learning

Inspect every source, assign business process/grain/key/owner role, profile basic quality and decide which source is required for each executive question without flattening incompatible facts.

3. Prerequisite / what you already know

CAP1 decision contract and C2-06 facts/grains/source-contract thinking.

4. Picture it in your head

> **Mental model**
>
> Imagine this as a visible path: Decision -> Source inventory -> Grain -> Role -> Quality risk -> Acquire/skip. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Source-of-truth role: The governed purpose for which a source/field is authoritative. Profiling: Initial examination of rows, columns, types, ranges, nulls, uniqueness and distributions. Fact-to-fact fan-out: Unintended multiplication when different business-process facts are joined at incompatible grains.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Read source_contracts.json before loading. Record grain/key/process for sales, marketing, targets, customer health and support; customers/products are dimensions.
2. Compare raw row/control expectations in control_totals.json with what you actually load.
3. Profile uniqueness/nulls/date span and known traps. Preserve raw copies; do not 'clean' evidence before documenting it.
4. Map each analytical question to the minimum needed facts. Revenue/margin can use sales; acquisition efficiency may combine region-period aggregates, not line-level direct joins; support remains its own process.
5. Mark sources not needed for a given question and why. More data is not automatically better analysis.

8. What you should see / be able to say

Expected result

A source/grain matrix plus acquisition plan that predicts where joins must be aggregate-to-compatible-grain or remain separate.

9. Why it worked

Reasoning

You assigned semantics before relationships, protecting the model from accidental fact multiplication. Professional upgrade: Production acquisition also records schema version, arrival/freshness, source owner, privacy classification and failure behavior. CAP2 asks you to model those fields even in synthetic work.

10. Common beginner mistakes

1. Load first and infer grain later. | 2. Use customer_id to directly join CustomerHealth to Sales lines for all analysis. | 3. Treat regional targets as a dimension because it is small. | 4. Delete support duplicate before recording the observed quality issue.

11. If you get stuck - recovery ladder

Recovery

For each file finish the sentence 'one row represents...' and identify the business process. If two answers differ, do not assume a direct join is safe. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Create source inventory for all seven business-data files, including row count, grain/key, purpose, date span, null/duplicate finding and required CAP question.

13. Validate your result

Validation

Reconcile all independent raw controls; explicitly detect the support duplicate and missing priority; show why Target and Marketing cannot be copied onto every Sales row.

14. Teach it back in your own words

Explain without reading

Explain why two tables sharing Region and Month can still require careful aggregation before comparison.

15. Protected review + changed fresh retry

Attempt protection

Save your source inventory. Changed retry: imagine Marketing changes from Month-Region-Channel to Week-Region-Campaign; document downstream impacts without modifying raw data.

16. Evidence / mastery + exact next route

Mastery evidence

Source/grain register + profile outputs + raw-control reconciliation + acquisition/impact notes + explain-back. Meridian data ZIP + source_contracts.json + control_totals.json. Then continue to CAP3.

CAP3 - SQL extraction: produce and validate analytical dataset

1. Why it matters

A query that runs is not enough. CAP3 proves you can create maintainable analytical SQL whose output grain and totals survive joins, filtering and changed questions.

2. What you are learning

Design explicit grain-safe SQL using CTEs/window logic only where useful; avoid SELECT * and many-to-many shortcuts; save row/key/financial checks beside the extraction.

3. Prerequisite / what you already know

C2-02 Advanced SQL and CAP2 source grains. PostgreSQL execution is preferred when available but precise SQL design is still required.

4. Picture it in your head

> **Mental model**
>
> Imagine this as a visible path: Question -> Declare output gra... -> SQL -> Join safely -> Window/CTE -> Validate. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Extraction grain: What one row of the SQL output represents. Control query: Independent query used to validate row count, uniqueness, totals or boundaries. Window function: Calculation across related rows without collapsing them to one group row.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. State output grain first: one row per Month-Region for an executive trend table.
2. Aggregate sales lines to Month-Region before comparing with Month-Region target. Join customer region only through the validated customer dimension relationship.
3. Calculate revenue, cost and gross margin explicitly. Use CTEs to make stage/grain clear; use window logic only if growth/ranking needs it.
4. Create separate support or health extracts at compatible analytical grains instead of joining raw Ticket/Customer-Month facts to Sales lines.
5. Write independent controls: raw sales revenue/margin totals, output key uniqueness, date boundaries and target totals before/after comparison.

8. What you should see / be able to say

Expected result

SQL designs/outputs with declared grain and saved validation that cannot silently multiply the business facts.

9. Why it worked

Reasoning

You reduced each fact to a compatible analytical grain before comparison and checked results independently. Professional upgrade: For production, parameterize date bounds, separate transformation layers, add tests and query-performance evidence. Do not claim production execution if you only authored local SQL.

10. Common beginner mistakes

1. Join Target directly to Sales line then SUM(Target). | 2. Join Support tickets to Sales via customer and aggregate both. | 3. Use DISTINCT to hide a fan-out symptom. | 4. CTEs added only to sound advanced, without clear grain transitions.

11. If you get stuck - recovery ladder

Recovery

Print row counts and uniqueness after every join. If count grows, explain whether growth is intended by output grain before proceeding. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Write three maintainable extracts: Month-Region sales/target, Customer-Month health, and Support Month-Region summary. Include one window-based analysis justified by a question.

13. Validate your result

Validation

Reconcile sales revenue/cost/margin to raw control totals and prove each final extraction key matches declared grain.

14. Teach it back in your own words

Explain without reading

Explain why DISTINCT can hide a modeling problem rather than solve it.

15. Protected review + changed fresh retry

Attempt protection

Save SQL plus validation first. Changed retry: request changes from Month-Region to Quarter-Segment; modify output grain without copying a target value to customer-level rows.

16. Evidence / mastery + exact next route

Mastery evidence

SQL files/notebook + grain comments + control queries/results + one failure diagnosis + explain-back. Optional schema_and_seed.sql supports PostgreSQL-style execution; raw-control JSON remains independent check. Then continue to CAP4.

CAP4 - Transformation: choose SQL, Power Query or Python appropriately

1. Why it matters

Using the most advanced tool is not senior behavior. The right tool depends on where data already lives, data shape, repeatability, deployment ownership and who must maintain the result.

2. What you are learning

Choose SQL, Power Query or Python for each transformation; normalize CustomerHealth JSON, handle support quality traps, preserve raw evidence and document repeatable transformation contracts.

3. Prerequisite / what you already know

C2-01 Power Query, C2-05 Python automation, CAP2-3 grains/SQL.

4. Picture it in your head

Mental model

Imagine this as a visible path: Data shape -> Ownership -> Repeatability -> Tool choice -> Transform -> Validate. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Tool boundary: Where responsibility for a transformation belongs and who can operate it. Idempotent/repeatable transform: Transformation that can be rerun predictably without duplicated/corrupted business output. Quarantine: Keep invalid records separately for investigation instead of silently treating them as trusted.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. CustomerHealth JSON is semi-structured Customer-Month data. Python or Power Query can normalize it; choose based on repeatability/ownership and prove output object count/key.
2. Support has one exact duplicate and one blank priority. Detect first; decide dedupe/quarantine/impute policy separately; never silently fill priority without business rule.
3. Sales/Target aggregation close to a relational warehouse may belong in SQL; report-only shaping owned by a BI team may fit Power Query. Explain the boundary.
4. Preserve original raw files and write transformation log: input -> operation -> reason -> output grain -> validation.
5. Inject one changed field/null/schema condition and show whether transform fails, quarantines or adapts according to the contract.

8. What you should see / be able to say

Expected result

A transformation plan where each tool has a maintainability/ownership reason and every output can be validated/replayed.

9. Why it worked

Reasoning

Tool choice follows system context and operating responsibility, not prestige. Professional upgrade: At larger scale, transformations may move to warehouse/lakehouse/orchestrated pipelines. CAP4 still expects the same contracts, raw preservation, validation and ownership reasoning.

10. Common beginner mistakes

1. Python for every file because automation is advanced. | 2. Power Query silently removes duplicate without evidence. | 3. SQL transforms JSON only because 'SQL is more professional'. | 4. No raw-to-curated transformation record.

11. If you get stuck - recovery ladder

Recovery

Score each task on source location, shape, reuse, scale, deployment owner, learner/team skills and failure/replay need. Choose the simplest governed fit. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Normalize health JSON and clean Support using your chosen tools; create a one-page tool-choice matrix for five transformation tasks.

13. Validate your result

Validation

Health object count/key, support duplicate/missing-priority controls and post-transform row/status counts are reproducible.

14. Teach it back in your own words

Explain without reading

Explain why a simpler Power Query transform can be more professional than Python in some BI-owned workflows.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry: source expands to 50 daily JSON files and another team must operate it. Re-evaluate the tool boundary.

16. Evidence / mastery + exact next route

Mastery evidence

Tool-choice matrix + transformation artifact + exception log + before/after controls + explain-back. Meridian CustomerHealth/support sources; prior C2-01/C2-05 patterns may be reused as techniques, not answers. Then continue to CAP5.

CAP5 - Modeling: build the semantic/business model

1. Why it matters

The capstone intentionally contains Sales, Marketing, Targets, Health and Support at incompatible grains. A single giant table would make the report easy to start and hard to trust.

2. What you are learning

Design a star-style model with separate facts, conformed dimensions/date roles, explicit relationship cardinality/filter direction and KPI measure contracts.

3. Prerequisite / what you already know

C2-03 modeling, C2-04 DAX and C2-06 dimensional grain.

4. Picture it in your head

> **Mental model**
>
> Imagine this as a visible path: Processes -> Facts -> Dimensions -> Date roles -> Relationships -> Measures. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Conformed dimension: Shared descriptive dimension consistently filtering compatible facts. Role-playing date: Same Date dimension used for different business date roles through separate relationships/logic. Semantic contract: Documented meaning/grain/filter behavior for model objects/measures.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. DimCustomer, DimProduct and DimDate provide governed descriptions. Add Region/Channel dimensions only if they simplify and remain consistent.
2. FactSales grain one invoice line; FactMarketing Month-Region-Channel; FactTarget Month-Region; FactCustomerHealth Customer-Month; FactSupport Ticket.
3. Use compatible keys and single-direction relationships by default; do not create fact-to-fact many-to-many to force a visual.
4. Define date roles: invoice date, ticket date, health month, marketing month, target month. State which is active or handled in measures.
5. Write contracts for Total Revenue, Gross Margin, Margin %, Target Attainment and one Health/Support KPI; identify where a cross-fact measure requires compatible filter context.

8. What you should see / be able to say

Expected result

A model diagram and measure register where every fact's grain remains visible and filters do not create ambiguous paths.

9. Why it worked

Reasoning

You preserved business processes while letting conformed dimensions coordinate analysis. Professional upgrade: Production models also consider aggregation, incremental refresh, RLS, calculation groups and performance-but add them only if required by the decision/workload.

10. Common beginner mistakes

1. One combined table with repeated target/spend rows. | 2. Bidirectional filters everywhere. | 3. Dimension key not unique. | 4. Same unlabeled Date relationship assumed to mean every business event date.

11. If you get stuck - recovery ladder

Recovery

Write one-row sentence above every fact and draw dimensions around it. If a relationship connects two facts directly, stop and justify why. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Build or fully specify the Meridian model and create at least six measure contracts with filter/date assumptions.

13. Validate your result

Validation

Unique dimension keys, fact foreign-key coverage, relationship cardinality/direction, fact-row totals and cross-fact KPI test cases.

14. Teach it back in your own words

Explain without reading

Explain why keeping facts separate can make the model simpler to reason about even if it creates more tables.

15. Protected review + changed fresh retry

Attempt protection

Save model first. Changed retry: leadership adds weekly Channel targets; update model without attaching weekly targets to invoice lines.

16. Evidence / mastery + exact next route

Mastery evidence

Model diagram/PBIX screenshots + fact/grain register + relationship/date-role checks + measure contracts + explain-back. Use CAP2 source contracts and C2-03/C2-04 principles. Then continue to CAP6.

CAP6 - Analysis: calculate and investigate

1. Why it matters

The dataset contains intentionally suggestive patterns: West marketing spend is high, some customer health deteriorates, and support pressure may rise. Those patterns are invitations to investigate, not permission to invent a cause.

2. What you are learning

Investigate growth quality across revenue, margin, target attainment, acquisition efficiency, health and service risk; separate observations from hypotheses and test alternative explanations.

3. Prerequisite / what you already know

CAP1 decision contract, CAP3-5 trusted extracts/model and SBA4 evidence language.

4. Picture it in your head

Mental model

Imagine this as a visible path: Question -> Baseline -> Segment -> Driver -> Alternative -> Decision evidence. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Driver analysis: Investigation of factors associated with a metric change, not automatically causal proof. Segment comparison: Compare cohorts/regions/products/periods under consistent definitions. Triangulation: Use multiple independent measures/sources to strengthen or challenge an interpretation.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Start with revenue/margin/target trend versus other regions and prior periods. Record exact observed differences.
2. Compare marketing spend/acquisition efficiency at compatible Month-Region(-Channel) grain; do not attach spend to invoice lines.
3. Inspect CustomerHealth by segment over time and Support pressure separately; identify whether deterioration is concentrated in the same region/segment/period.
4. List alternative explanations: mix shift, seasonality, product mix, source completeness, support measurement changes, acquisition channel mix.
5. Conclude with calibrated language: what is observed, what is plausible, what is contradicted, and what evidence is still required before claiming cause.

8. What you should see / be able to say

Expected result

An analysis notebook/worksheet where the executive narrative can be traced to validated measures and explicit uncertainty.

9. Why it worked

Reasoning

Multiple sources inform the hypothesis while remaining semantically separate and independently checkable. Professional upgrade: If this were a real decision with material spend, design the next measurement/experiment needed to distinguish competing causes. Senior analysis can recommend a reversible action under uncertainty.

10. Common beginner mistakes

1. 'West spend caused bad health'. | 2. Only analyze the region you already expect to be weak. | 3. Use one absolute value without comparable baseline. | 4. Ignore data-quality trap because pattern looks plausible.

11. If you get stuck - recovery ladder

Recovery

For every claim write O, I or R. For each inference list one competing explanation and one check. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Answer the full healthy-vs-fragile question using at least five evidence lenses and identify one segment/region requiring action plus one that should not be overreacted to.

13. Validate your result

Validation

Every headline number links to a trusted measure/control; causal language is no stronger than evidence; opposite/alternative case is considered.

14. Teach it back in your own words

Explain without reading

Explain the difference between 'evidence points to' and 'data proves'.

15. Protected review + changed fresh retry

Attempt protection

Save analysis before protected review. Changed retry: remove Marketing data entirely. State which conclusions survive and which become weaker.

16. Evidence / mastery + exact next route

Mastery evidence

Analysis workbook/notebook + O/I/R register + segment comparisons + alternative-hypothesis table + explain-back. Meridian sources plus your validated model; no solved executive conclusion is supplied. Then continue to CAP7.

CAP7 - Power BI: create the decision-facing report

1. Why it matters

A technically correct model can still produce a weak executive experience. The report must help leadership decide what to do, not demonstrate how many visuals you know.

2. What you are learning

Design an executive report with decision-oriented KPI/trend/driver/exception views, accessible titles/units/status, purposeful interactions and documented security/freshness assumptions.

3. Prerequisite / what you already know

C2-04 reporting/storytelling and CAP5-6 model/analysis.

4. Picture it in your head

Mental model

Imagine this as a visible path: Audience -> KPI page -> Trend -> Drivers -> Exceptions -> Action. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Decision-facing report: Report structured around user decisions/actions rather than data-table inventory. Exception view: View highlighting cases that breach a rule/threshold and require attention. Security assumption: Documented assumption about who can see which report/model/data and why.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Page 1 Executive health: Revenue, Gross Margin %, Target Attainment plus short headline/selected period.
2. Page 2 Trend/drivers: Region/segment/channel views that explain where performance differs without mixing incompatible facts.
3. Page 3 Customer/service risk: Health/Support patterns at valid grains with caveat if causal link is unproven.
4. Page 4 Action/exceptions only if it helps: priority region/segment/action with owner/metric/next checkpoint.
5. Test navigation/filter interactions, mobile/accessibility needs, units, titles and security/freshness notes. Remove visuals that do not support a decision.

8. What you should see / be able to say

Expected result

A concise report where an executive can identify healthy/fragile growth and next action without discovering grain/model caveats by accident.

9. Why it worked

Reasoning

The report surfaces the decision and supporting evidence while model complexity stays behind the scenes. Professional upgrade: Production delivery adds workspace/app/deployment/refresh/security operations from C2-07. CAP7 requires those assumptions documented even if your local project cannot execute the tenant steps.

10. Common beginner mistakes

1. One page with 15 visuals. | 2. Red/green without accessible labels. | 3. Marketing spend appears to filter line-level revenue incorrectly. | 4. Hidden slicer or interaction changes KPI meaning unexpectedly.

11. If you get stuck - recovery ladder

Recovery

For each visual answer: which decision/question does this help? If no answer, remove it. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Build/report-spec the complete Meridian executive report and run an interaction/accessibility/security/freshness checklist.

13. Validate your result

Validation

KPI totals reconcile to CAP8 controls; filter behavior matches measure contract; titles/units/accessibility are clear; report does not expose unsupported causal claims.

14. Teach it back in your own words

Explain without reading

Explain why report polish cannot repair a wrong model or unvalidated metric.

15. Protected review + changed fresh retry

Attempt protection

Save report evidence first. Changed retry: CFO asks for a one-page monthly review version; simplify without removing the material limitation.

16. Evidence / mastery + exact next route

Mastery evidence

PBIX/report screenshots or detailed report spec + interaction/accessibility checklist + KPI reconciliation + explain-back. Internal course sufficient; use C2-04/C2-07 references only to repair a specific weak area. Then continue to CAP8.

CAP8 - Validation: reconcile important numbers independently

1. Why it matters

Capstone credibility depends on proof that the story is built on the same business facts from source to report. CAP8 is where attractive-but-wrong work gets blocked.

2. What you are learning

Create at least eight independent controls across raw rows, keys, referential/cardinality, financial totals, target grain, date boundaries, quality traps and report/model output; define blocking versus warning failures.

3. Prerequisite / what you already know

C2-00 validation, C2-06 quality, CAP2-7 artifacts.

4. Picture it in your head

Mental model

Imagine this as a visible path: Raw control -> Transform control -> Model control -> Report control -> Failure -> Release. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Reconciliation: Independent comparison showing two representations of the same business quantity agree within an explicit rule. Blocking control: Failure severe enough to stop publication/decision use. Control evidence: Saved expected, observed, status, timestamp and owner/action-not just 'checked'.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Raw: verify 576 Sales rows and raw revenue/cost/gross-margin totals from control_totals.json.
2. Quality: Customer/Product key uniqueness, Support exact duplicate count 1 and blank Priority count 1 before governed handling.
3. Model: fact rows and foreign keys; Target/Marketing grains not multiplied; Health Customer-Month uniqueness.
4. Date/period: compare first/last dates and month counts to intended analysis period; make future/incomplete periods explicit.
5. Report: selected no-filter executive Revenue/Margin controls reconcile to trusted model/extract. Record each check as PASS/WARN/FAIL and block release for critical unresolved failures.

8. What you should see / be able to say

Expected result

A validation log that can tell another analyst exactly why the current capstone is safe-or not safe-to release.

9. Why it worked

Reasoning

Controls are independent and distributed across the data path, so failure can be localized. Professional upgrade: In production, controls become automated tests/monitoring/release gates. CAP8's evidence format deliberately transfers to that engineering practice.

10. Common beginner mistakes

1. One grand-total check only. | 2. Use the same DAX measure to 'validate' itself. | 3. Delete quality traps then report zero issues. | 4. Show FAIL in workbook but still publish final recommendation.

11. If you get stuck - recovery ladder

Recovery

List risks by layer: arrival/raw -> schema/key -> transform -> model/join -> business total -> report/security/freshness. Add one independent control per risk. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Implement at least 10 controls including all supplied raw-control totals, support traps, target/marketing grain and final report KPIs. Inject one failed control and demonstrate publication block.

13. Validate your result

Validation

Each check has rule, observed, status, severity, evidence ID and owner/action; failed critical check prevents release.

14. Teach it back in your own words

Explain without reading

Explain why an independent control should not reuse exactly the same logic as the metric it validates.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry: support duplicate is fixed upstream but a new Product key becomes missing; route the new failure without rewriting the whole QA system.

16. Evidence / mastery + exact next route

Mastery evidence

Validation log + raw/model/report reconciliation + injected-failure block + repair/retest evidence + explain-back. control_totals.json is an independent baseline; do not edit it. Then continue to CAP9.

CAP9 - Executive communication: recommendations and limitations

1. Why it matters

The capstone is not complete when the dashboard is complete. Leadership needs a concise recommendation whose strength matches the evidence.

2. What you are learning

Write a decision-first executive brief with headline, quantified evidence, implication, recommendation, owner/timing, limitation/uncertainty and next evidence; keep method detail in appendix unless it changes the decision.

3. Prerequisite / what you already know

SBA4-SBA7 plus CAP6-8 validated findings.

4. Picture it in your head

> **Mental model**
>
> Imagine this as a visible path: Headline -> Evidence -> Implication -> Action -> Owner -> Limitation/next. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Decision ask: Specific approval/action/resource choice requested from the audience. Material limitation: Uncertainty or data limitation capable of changing interpretation/action. Guardrail: Metric/condition preventing the recommended action from causing an unacceptable side effect.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Headline: say where growth appears healthy/fragile and the decision you recommend-without claiming unsupported cause.
2. Evidence: choose 2-3 validated numbers spanning growth quality rather than metric dumping.
3. Implication: what business risk/opportunity exists next quarter if no action changes?
4. Recommendation: specific reversible action/owner/timing plus success metric and guardrail.
5. Limitation/next evidence: name the strongest uncertainty and the next measurement/checkpoint that could change the recommendation.

8. What you should see / be able to say

Expected result

An executive can approve/reject/modify the action in under a minute and knows what uncertainty remains.

9. Why it worked

Reasoning

The brief follows decision -> evidence -> action rather than analysis chronology. Professional upgrade: Tailor depth to audience while keeping a stable evidence appendix and decision log. If two executives have different decisions, they may need different summaries using the same governed facts.

10. Common beginner mistakes

1. Start with SQL/cleaning. | 2. Claim employer revenue impact from a synthetic project. | 3. Recommendation says only 'monitor West'. | 4. Caveat hidden in appendix even though it could reverse the recommendation.

11. If you get stuck - recovery ladder

Recovery

Use six lines: headline/decision; evidence 1; evidence 2; implication; action/owner; limitation + next checkpoint. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Write <=120 words and then <=80 words for the Meridian recommendation while preserving the material limitation and decision ask.

13. Validate your result

Validation

All numbers trace to CAP8; causal language is calibrated; action has owner/timing/measure/guardrail; synthetic nature is not misrepresented.

14. Teach it back in your own words

Explain without reading

Explain why a good executive summary can be shorter than one dashboard page but more useful for a decision.

15. Protected review + changed fresh retry

Attempt protection

Save both versions first. Changed retry: leadership will fund only half the recommended intervention; revise action while keeping evidence unchanged.

16. Evidence / mastery + exact next route

Mastery evidence

120-word + 80-word briefs + evidence references + recommendation/guardrail + explain-back. SBA5-SBA7 are the internal teacher references; no external source required. Then continue to CAP10.

CAP10 - Documentation: create an inheritable analysis package

1. Why it matters

A senior project should survive your absence. If another analyst cannot identify source grain, rerun transformations, reconcile results and understand limitations, the capstone is a demo rather than an operational analysis package.

2. What you are learning

Create handover-quality source/grain dictionary, metric contracts, transformation log, source-to-report lineage, rerun/refresh instructions, known issues, validation log and ownership assumptions.

3. Prerequisite / what you already know

C2-05 reproducibility, C2-06 lineage/quality, C2-07 operations and CAP1-9.

4. Picture it in your head

Mental model

Imagine this as a visible path: Source dictionary -> Metric contract -> Transformation -> Lineage -> Runbook -> Known issues. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Runbook: Step-by-step operating/recovery instructions for repeating or restoring an analytical process. Data dictionary: Document describing fields/tables, grain, meaning, type and ownership. Known-issues register: Explicit list of unresolved limitations/risks, impact, owner and workaround/next action.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. README: decision, audience, folder structure, software/tool assumptions and how to reproduce from raw sources.
2. Source/grain dictionary and metric contracts: one-row meaning, keys, dates, formulas/filter rules, owners/approvals.
3. Transformation/lineage: raw -> cleaned/extract -> model -> measure -> report, including the support-trap handling and health normalization.
4. Validation/runbook: exact order to run, expected controls, failure/retry behavior, optional Power BI refresh/deployment assumptions.
5. Known issues/production gaps: synthetic/local limits, security/credentials, monitoring, scale, data freshness and any unresolved definition.

8. What you should see / be able to say

Expected result

A handover package another competent analyst can operate/review without a meeting to rediscover basic assumptions.

9. Why it worked

Reasoning

Business meaning, technical flow and validation are all documented together. Professional upgrade: Production documentation should live with/version alongside the solution and be updated by change. Avoid confidential evidence in a public portfolio.

10. Common beginner mistakes

1. README says only 'open PBIX'. | 2. Screenshots without exact definitions/paths. | 3. Known limitation omitted to make portfolio look perfect. | 4. Documentation describes final state but not rerun/failure path.

11. If you get stuck - recovery ladder

Recovery

If another analyst would have to ask you 'what does this row/metric mean?' or 'what do I run first?', add that answer to the package. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Build the full Meridian handover index and have a cold-start checklist that another person could follow from raw ZIP to validated executive output.

13. Validate your result

Validation

Every major artifact is referenced; paths/names are current; metric/source grains match CAP1-8; known issues and production gaps are explicit.

14. Teach it back in your own words

Explain without reading

Explain why documentation is part of correctness, not administrative cleanup after correctness.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry: rename/move the project root and hand it to a fresh environment; identify every hard-coded assumption that breaks.

16. Evidence / mastery + exact next route

Mastery evidence

README + dictionaries/contracts + lineage + runbook + known-issues register + cold-start test + explain-back. C2-05/C2-06 documentation patterns transfer directly. Then continue to CAP11.

CAP11 - Technical defense: explain SQL, model, DAX and validation decisions

1. Why it matters

Interviewers and reviewers change the question. Memorized code fragments collapse when asked 'why this grain?', 'what would fail?', or 'why not the other architecture?'. CAP11 turns the project into defendable reasoning.

2. What you are learning

Prepare evidence-linked answers for source/grain, SQL joins/windows, model relationships/date roles, DAX/filter context, tool choice, validation/failure recovery and production-scale trade-offs.

3. Prerequisite / what you already know

CAP1-10 and the technical modules.

4. Picture it in your head

Mental model

Imagine this as a visible path: Decision -> Technical choice -> Evidence -> Alternative -> Failure -> Production change. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Technical defense: Evidence-based explanation of why a design is correct/appropriate and what trade-offs it has. Counterfactual: Alternative design/explanation considered and rejected for a reason. Production reflection: Clear separation between what was actually built and what would change in a real production environment.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Why are Marketing and Target separate facts? Answer with grain and fan-out risk, not 'star schema best practice'.
2. How do you know SQL did not multiply revenue? Point to row/key/financial controls.
3. Why is this DAX a measure not calculated column? Explain evaluation context/business need.
4. Why Python/Power Query/SQL for the chosen transformation? Explain ownership/repeatability/data shape.
5. What failed? Use supplied support trap or your injected validation failure and show block/recovery evidence.
6. What changes at 10x scale/production? Separate actual project from proposed orchestration, security, monitoring, incremental processing or warehouse changes.

8. What you should see / be able to say

Expected result

Short answers tied to saved project evidence and able to survive a changed follow-up.

9. Why it worked

Reasoning

You explain reasoning/controls, not tool-name prestige. Professional upgrade: Keep answers concise first; hold deeper evidence in an appendix. State 'I would' for production extensions you did not execute.

10. Common beginner mistakes

1. Quote DAX formula before business meaning. | 2. Say 'because best practice'. | 3. Claim production experience from local project. | 4. Hide failure to make answer sound stronger.

11. If you get stuck - recovery ladder

Recovery

For each answer use Decision -> Choice -> Evidence -> Alternative -> Limitation/Production change. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Create 12-question technical defense pack across SQL, model, DAX/report, automation, quality, cloud/operations and production reflection. Answer aloud once if comfortable.

13. Validate your result

Validation

Every answer points to an evidence ID/file and includes at least one validation/limitation/alternative.

14. Teach it back in your own words

Explain without reading

Explain why a failure you detected/recovered can make an interview story stronger than pretending nothing failed.

15. Protected review + changed fresh retry

Attempt protection

Save first defense. Changed retry: answer the same competencies with different wording/questions and no notes.

16. Evidence / mastery + exact next route

Mastery evidence

12-question defense pack + evidence links + optional voice notes + changed follow-up responses + explain-back. Use C2_09_SENIOR_BI_INTERVIEW_COMPANION_RC1 after the first attempt, not before. Then continue to CAP12.

CAP12 - Senior Analyst interview preparation

1. Why it matters

A long course is not an interview answer. You need 2-3 compact stories that prove specific capabilities without turning a synthetic project into fake employment experience.

2. What you are learning

Convert capstone/course evidence into truthful Situation-Task-Action-Evidence/Learning stories; map job requirements to proof; prepare changed follow-ups and identify gaps you cannot yet claim.

3. Prerequisite / what you already know

CAP11 technical defense and C2-08 communication.

4. Picture it in your head

Mental model

Imagine this as a visible path: Job requirement -> Evidence vault -> Story -> Truth check -> Follow-up -> Gap. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

STAR/S-T-A-R: Interview structure: Situation, Task, Action, Result; this course adds evidence/learning explicitly. Evidence vault: Index of saved artifacts that can substantiate claims. Truth boundary: Line separating what you actually did from proposed/learned/production extensions.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Situation: synthetic Meridian leadership needed trustworthy growth-quality analysis with multiple source grains.
2. Task: you owned validation/reconciliation before executive recommendation.
3. Action: state the actual controls you implemented-raw totals, keys, fact grain, quality traps, model/report reconciliation.
4. Evidence/result: 'My project validation caught/contained the supplied duplicate/missing priority and reconciled revenue/margin to raw controls.' Do not say employer money was saved.
5. Learning/production reflection: what would you automate/monitor in a real environment and what remains unproven. Prepare follow-up: 'How did you know it was correct?'

8. What you should see / be able to say

Expected result

At least three truthful stories with artifact references and no invented tenure/impact.

9. Why it worked

Reasoning

Claims are grounded in specific actions/evidence rather than generic 'I know SQL/Power BI'. Professional upgrade: Target roles vary. Customize which evidence stories you choose, not the truth of the evidence. Keep a gap list for requirements you cannot yet defend.

10. Common beginner mistakes

1. Synthetic project described as employer production system. | 2. Invent revenue/salary/user impact. | 3. Story lists tools but no decision/validation. | 4. Memorize one script and fail when follow-up changes.

11. If you get stuck - recovery ladder

Recovery

If a sentence cannot point to saved evidence, downgrade the claim or move it to 'I would do in production'. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Create three stories: technical quality, stakeholder/decision, and automation/modeling. Map each to a target Senior/BI requirement and one changed follow-up.

13. Validate your result

Validation

Truth check passes every claim; evidence IDs exist; story fits 60-90 seconds; result language distinguishes project evidence from workplace impact.

14. Teach it back in your own words

Explain without reading

Explain the difference between 'I built this in my project' and 'I operated this in production'.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry: answer a job description that emphasizes governance/refresh instead of SQL; choose different evidence without inventing it.

16. Evidence / mastery + exact next route

Mastery evidence

3+ interview stories + job-requirement mapping + truth-check column + changed follow-ups + explain-back. Use interview companion and Promotion Readiness Workbook after first story drafts. Then continue to CAP13.

CAP13 - Internal promotion case

1. Why it matters

Completing a course does not mean an employer owes a promotion. A responsible internal case demonstrates capability/ownership using sanitized evidence and asks for a concrete next step while acknowledging gaps and organizational constraints.

2. What you are learning

Build an evidence-backed promotion case across technical ownership, analytical judgment, communication, automation, quality, documentation/mentoring and business value; separate course/project/workplace evidence.

3. Prerequisite / what you already know

CAP1-12 and C2_08 SBA10-12. Workplace evidence is optional; never fabricate it.

4. Picture it in your head

Mental model

Imagine this as a visible path: Target capability -> Evidence matrix -> Business value -> Ownership -> Gaps -> Specific ask. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Promotion case: Evidence-backed argument for increased scope/title/readiness, not a guarantee. Evidence strength: This course's 0-4 scale distinguishing none, course-only, independent project, sanitized workplace and repeated ownership. Scope request: Specific responsibility/opportunity requested even if formal title change is not immediately available.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Choose target capability/role and list the 8-12 capabilities that role actually requires in your organization.
2. For each capability, record strongest evidence ID and evidence strength. Course-only learning is weaker than independent project/workplace ownership.
3. Write why now using observable readiness/scope, not 'I finished a certificate'.
4. State known gaps and how you will close them. Do not hide lack of production exposure.
5. Make a specific ask: expanded ownership, review responsibility, project lead, promotion discussion or measurable 90-day scope-depending on organizational context.

8. What you should see / be able to say

Expected result

A promotion-readiness document that is credible even if the manager says 'not yet' or title budgets are constrained.

9. Why it worked

Reasoning

Capability evidence and employer decision are kept separate. Professional upgrade: Salary/title decisions depend on employer structure, market, timing and negotiation. This course measures capability evidence only and never promises promotion outcomes.

10. Common beginner mistakes

1. Course completion = promotion entitlement. | 2. Confidential internal data copied into portfolio/workbook. | 3. Inflate synthetic project into workplace impact. | 4. No specific ask other than 'promote me'.

11. If you get stuck - recovery ladder

Recovery

If you have no workplace evidence yet, build an internal readiness/gap plan rather than pretending. Use course/project proof and request opportunities to create workplace evidence. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Complete the Promotion Readiness Workbook competency matrix, evidence vault and promotion case using only evidence you truly possess.

13. Validate your result

Validation

Every claim has evidence ID/strength; confidentiality check passes; known gaps are visible; requested next step is specific and reasonable.

14. Teach it back in your own words

Explain without reading

Explain why being ready for more responsibility and receiving a promotion are related but not identical.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry: manager says no title change this cycle but offers ownership of one cross-functional BI product. Write a 90-day evidence-building plan.

16. Evidence / mastery + exact next route

Mastery evidence

Competency matrix + evidence vault + promotion case + confidentiality check + 90-day plan + explain-back. Use C2_09_PROMOTION_AND_JOB_SWITCH_READINESS_WORKBOOK_RC1. Then continue to CAP14.

CAP14 - External job-switch preparation

1. Why it matters

A polished capstone is useful only if you can map it to a real role without misrepresenting production experience. CAP14 turns course evidence into a job-switch packet and explicitly records what must still be learned or proven.

2. What you are learning

Select target roles, map must-have requirements to saved proof/gaps, create a concise portfolio story, prepare production-reflection answers and choose Apply/Maybe/Skip based on evidence rather than fear or wishful thinking.

3. Prerequisite / what you already know

CAP11-13 and entire C2A evidence vault.

4. Picture it in your head

Mental model

Imagine this as a visible path: Target role -> Evidence requireme... -> Portfolio story -> Truth boundary -> Production gaps -> Apply/skip. Keep the stages separate. If the result becomes wrong, walk backward one stage at a time until you find where the meaning, rule, grain, relationship, or control changed.

5. New words before we use them

Evidence match: Specific saved proof that demonstrates a stated job requirement. Production gap: Capability not actually demonstrated at production scale/security/reliability even if concept is understood. Apply threshold: Practical rule for applying based on role evidence and learnability, not a claim you meet every bullet.

6. Watch / visual source

INTERNAL TEACHING ROUTE. No external video is required for this lesson. Use the mental model and worked demonstration below as the canonical teaching route. This capstone stays independent, so no solved walkthrough is inserted.

7. Follow with me once

Teacher demonstration - do this once with me, then close the example before your fresh task.

1. Target one role family at a time: Senior Data Analyst / BI Analyst / Power BI Analyst. Separate C2A role path from Data Engineering route.
2. Map requirements: advanced SQL, modeling/DAX, Power BI, automation, quality, stakeholder decisions, documentation; link each to project/work evidence.
3. Write portfolio story: problem -> data/grains -> decisions/tooling -> validations -> findings/recommendation -> limitations -> production changes.
4. Production reflection: security, deployment, monitoring, orchestration, scale, credentials, SLAs and real business impact are not claimed unless actually evidenced.
5. Classify requirement as Proof / Learnable gap / Major gap. Apply when core role evidence is strong enough; Maybe when one or two learnable gaps remain; Skip when critical must-have evidence is absent.

8. What you should see / be able to say

Expected result

A job-switch evidence matrix and portfolio narrative that you can defend under technical and behavioral follow-ups.

9. Why it worked

Reasoning

You market real capability while keeping production claims inside the truth boundary. Professional upgrade: Job markets change. Re-check target postings when you apply, but keep the evidence system stable. For Data Engineering transition, use the C2B/C3 route or the explicit C2A readiness check-not title inflation.

10. Common beginner mistakes

1. Resume says production pipeline after local notebook only. | 2. Apply to Data Engineer role using Senior BI capstone as if it proves DE job readiness. | 3. List every tool learned without evidence. | 4. Skip every role because one optional bullet is missing.

11. If you get stuck - recovery ladder

Recovery

For each requirement ask: what saved artifact proves this? If none, label gap. Do not replace the gap with confidence language. Preserve your first attempt before review.

12. Now you do a small version alone

Fresh independent task

Create two target-role evidence matrices and one concise public-safe Meridian portfolio README/story. Add production-gap plan and five interview follow-ups.

13. Validate your result

Validation

Every claimed skill maps to evidence; no confidential/synthetic-project misrepresentation; role classification/next action is explicit.

14. Teach it back in your own words

Explain without reading

Explain why a production-reflection section increases credibility rather than making the portfolio weaker.

15. Protected review + changed fresh retry

Attempt protection

Attempt first. Changed retry: target posting removes Power BI emphasis and adds stronger SQL/Python. Re-rank evidence and gaps without rewriting your history.

16. Evidence / mastery + exact next route

Mastery evidence

Job requirement matrices + public-safe portfolio story + production-gap plan + apply decision + explain-back. Promotion/Job-Switch Workbook + DE Readiness Check; PL-300 remains separate supplementary certification.

C2-09 core mastery checklist + quick reference

Use only after CAP1-CAP14 first pass

FOURTEEN QUESTIONS

CAP1 Is the decision contract explicit before calculation? | CAP2 Can every source grain/process/key/quality risk be explained? | CAP3 Do SQL outputs retain declared grain and independent controls? | CAP4 Does tool choice follow ownership/shape/repeatability? | CAP5 Are incompatible facts separate with safe relationships/date roles? | CAP6 Are observation/hypothesis/recommendation separated? | CAP7 Does report serve a decision and reconcile? | CAP8 Do critical controls block release? | CAP9 Can leadership act from a concise, truthful brief? | CAP10 Can another analyst inherit/rerun it? | CAP11 Can you defend choices/failures/alternatives? | CAP12 Are interview stories truthful and evidence-linked? | CAP13 Is promotion readiness evidence-backed without entitlement? | CAP14 Does job-switch story separate project proof from production gaps?

BLOCK CORE PASS IF

Fact/grain fan-out; unresolved critical validation; silent definition conflict; causal overclaim; report totals not reconciled; protected review opened before attempt; synthetic project described as employer/production impact; handover cannot reproduce; Final Gate critical failure.

SEPARATE STATUS SYSTEMS

Core C2A capability: CAP + Final Gate. Promotion/job switch: evidence readiness only. PL-300: optional certification prep. DE transition: direct route normally C2B; if using C2A evidence, use the separate C2A-to-DE readiness check. None should be silently merged.

MERIDIAN RAW CONTROL

Frozen capstone ZIP remains byte-identical to authoritative baseline, CRC PASS, all raw controls pass; deliberate support traps remain one exact duplicate and one missing Priority.

C2-09 internal/source route register

External media is not required for the core capstone

CAP1 - C2-08 SBA1/SBA4/SBA7

Decision discovery, evidence language and recommendation contracts.

CAP2 - C2-06 DWF4/DWF12 + Meridian source_contracts.json

Grain, source roles and quality.

CAP3 - C2-02 Advanced SQL

Grain-safe extraction, CTE/window, validation.

CAP4 - C2-01 Power Query + C2-05 Python

Tool choice, repeatability, exception handling.

CAP5 - C2-03 + C2-04

Semantic model, relationships, DAX contracts.

CAP6 - C2-08 SBA4/SBA6

Observation/inference, uncertainty and alternative explanations.

CAP7 - C2-04 + C2-07

Decision report plus lifecycle/security/freshness assumptions.

CAP8 - C2-00 + C2-06 DWF12

Independent reconciliation and blocking controls.

CAP9 - C2-08 SBA5-SBA7

Executive summary and decision-ready recommendation.

CAP10 - C2-05 PYA11 + C2-06 DWF11

Reproducibility, lineage, runbook, known issues.

CAP11 - C2-09 Senior/BI Interview Companion

Technical defense after first independent defense draft.

CAP12 - Promotion/Job-Switch Readiness Workbook

Evidence stories and truth checks.

CAP13 - Promotion/Job-Switch Readiness Workbook

Capability/evidence strength and internal case.

CAP14 - Promotion/Job-Switch Readiness Workbook + DE Readiness

Target-role evidence and production-gap plan.

IMPORTANT

No CAP source link supplies the Meridian answer. Re-open prior teaching only to repair a competency. PL-300 official sources are isolated in the supplementary book.